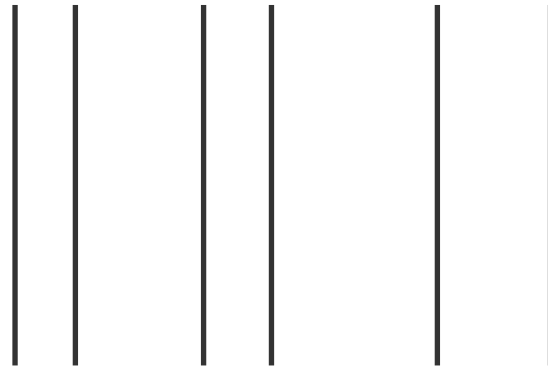


Four ways to ask if there is a calcium kernel

the kernel = the calcium bump caused by ONE action potential

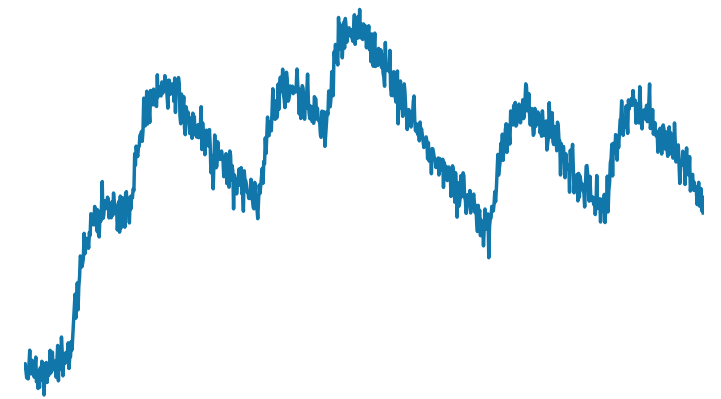
action potentials
(we MEASURE these)



the KERNEL — unknown



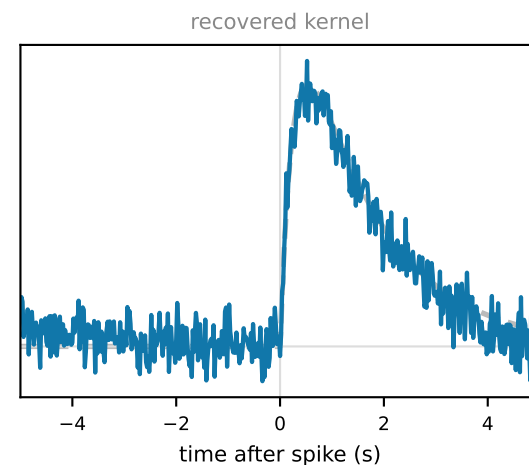
calcium signal
(we MEASURE this)



FREE-VECTOR

Mathematically undo the mixing (spikes $\rightarrow$ calcium).
No shape assumed.

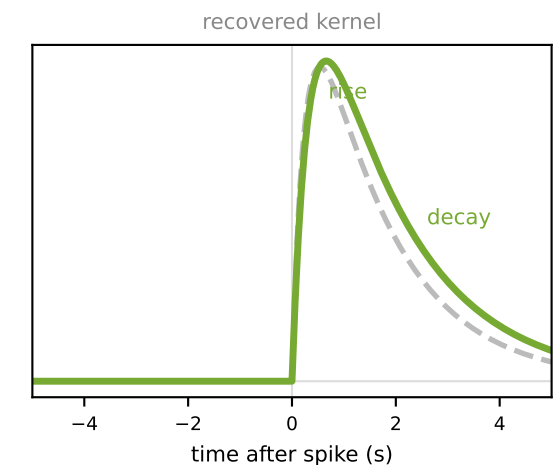
- ✓ recovers ANY shape
- ⚠ wiggly, baseline can drift



PARAMETRIC

Assume the shape (fast rise + slow decay); fit just a few numbers.

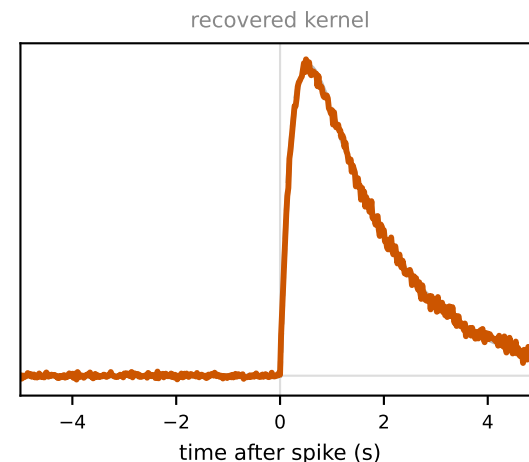
- ✓ clean, few knobs
- ⚠ wrong if the shape differs



SHAPED

Undo the mixing, but keep it behaving — smooth, flat baseline, no pre-spike bump.

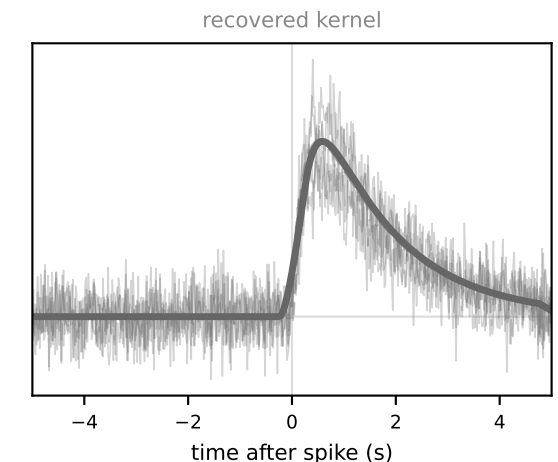
- ✓ flexible AND clean
- ⚠ penalties can bias the shape



STA

Snip the calcium around every spike and just average the snippets.

- ✓ simple, assumption-free
- ⚠ blurs when spikes crowd



If a real kernel exists, all four roughly AGREE. Disagreement is the diagnostic.